

Non-linear p-form gauge theories and their deformations

Master Degree in Physics — Final Dissertation

Candidate: Filippo Pesce

Supervisors: Prof. Kurt Lechner, Prof. Dmitri Sorokin

Dipartimento di Fisica e Astronomia “Galileo Galilei”
Università degli Studi di Padova

Academic Year 2025/2026

Studying QFTs via “deformations”

Starting from a “seed” theory described by a Lagrangian density $\mathcal{L}_0$, one way to deform the theory is to add an integrated local operator to the initial action

$$S_0 = \int d^d x \mathcal{L}_0(\phi, \partial_\mu \phi, \dots)$$



$$S_\lambda = S_0 + \lambda \int d^d x \mathcal{O}(\phi, \partial_\mu \phi, \dots) + o(\lambda^2).$$

Equivalent point of view: a flow (differential) equation

$$\frac{\partial S_\lambda}{\partial \lambda} = \int d^d x \mathcal{O}^{(\lambda)}(\phi, \partial_\mu \phi, \dots), \quad S_{\lambda=0} = S_0.$$

This thesis focuses on deformations of (chiral) 4-form gauge theories in $d = 10$, higher-rank analogues of Maxwell theory that arise naturally in string theory.

Studying QFTs via “deformations”

Starting from a “seed” theory described by a Lagrangian density $\mathcal{L}_0$, one way to deform the theory is to add an integrated local operator to the initial action

$$S_0 = \int d^d x \mathcal{L}_0(\phi, \partial_\mu \phi, \dots)$$

↓

$$S_\lambda = S_0 + \lambda \int d^d x \mathcal{O}(\phi, \partial_\mu \phi, \dots) + o(\lambda^2).$$

Equivalent point of view: a flow (differential) equation

$$\frac{\partial S_\lambda}{\partial \lambda} = \int d^d x \mathcal{O}^{(\lambda)}(\phi, \partial_\mu \phi, \dots), \quad S_{\lambda=0} = S_0.$$

This thesis focuses on deformations of (chiral) 4-form gauge theories in $d = 10$, higher-rank analogues of Maxwell theory that arise naturally in string theory.

$T\bar{T}$ -deformations

Rather surprisingly, the so called $T\bar{T}$ -irrelevant deformation in $2d$ conformal field theories has gained much attention in the last decade

Zamolodchikov '04

$$\mathcal{O}^{(\lambda)} \equiv \text{Det}(T_{\mu\nu}^{(\lambda)}) = T\bar{T}^{(\lambda)}.$$

- $T_{\mu\nu}^{(\lambda)}$ is the (deformed) stress-energy tensor.

What is special about $T\bar{T}$ -deformations?

- Universality;
- Solvability;
- Preserve many properties like supersymmetry and integrability.

Natural question

Can we extend this definition in $d > 2$?

$T\bar{T}$ -deformations

Rather surprisingly, the so called $T\bar{T}$ -irrelevant deformation in $2d$ conformal field theories has gained much attention in the last decade

Zamolodchikov '04

$$\mathcal{O}^{(\lambda)} \equiv \text{Det}(T_{\mu\nu}^{(\lambda)}) = T\bar{T}^{(\lambda)}.$$

- $T_{\mu\nu}^{(\lambda)}$ is the (deformed) stress-energy tensor.

What is special about $T\bar{T}$ -deformations?

- Universality;
- Solvability;
- Preserve many properties like supersymmetry and integrability.

Natural question

Can we extend this definition in $d > 2$?

T $\bar{T}$ -deformations

Rather surprisingly, the so called T $\bar{T}$ -irrelevant deformation in $2d$ conformal field theories has gained much attention in the last decade

Zamolodchikov '04

$$\mathcal{O}^{(\lambda)} \equiv \text{Det}(T_{\mu\nu}^{(\lambda)}) = T\bar{T}^{(\lambda)}.$$

- $T_{\mu\nu}^{(\lambda)}$ is the (deformed) stress-energy tensor.

What is special about T $\bar{T}$ -deformations?

- Universality;
- Solvability;
- Preserve many properties like supersymmetry and integrability.

Natural question

Can we extend this definition in $d > 2$?

Non-linear electrodynamics

Novel higher-dimensional realizations of stress-tensor deformations were discovered within non-linear electrodynamics (NED)

$$S_{\text{NED}} = \int d^4x \mathcal{L}_{\text{NED}}(F_{\mu\nu} F^{\mu\nu}, F_{\mu\nu} F^{\star\mu\nu})$$

where $F_{\mu\nu}^{\star} \equiv \frac{1}{2} \varepsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}$ is the Hodge-dual of the Maxwell tensor $F_{\mu\nu}$ (in $d = 4$).

Important examples (both recover $\mathcal{L}_{\text{Max}} \equiv -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ for $\gamma, \lambda \rightarrow 0$):

ModMax theory, $\gamma \rightarrow$ dimensionless parameter

$$\mathcal{L}_{\text{ModMax}}^{(\gamma)} = -\frac{1}{4} \cosh(\gamma) F_{\mu\nu} F^{\mu\nu} + \frac{1}{4} \sinh(\gamma) \sqrt{(F_{\mu\nu} F^{\mu\nu})^2 + (F^{\mu\nu} F_{\mu\nu}^{\star})^2},$$

Born-Infeld theory, $\lambda \rightarrow$ dimensionful parameter

$$\mathcal{L}_{\text{BI}}^{(\lambda)} = \frac{\sqrt{-g}}{\lambda} - \frac{1}{\lambda} \sqrt{-\text{Det}(g_{\mu\nu} + \sqrt{\lambda} F_{\mu\nu})}.$$

ModMax and BI as $T\bar{T}$ -like deformations of 4d Maxwell's theory

$$\begin{array}{ccc}
 \mathcal{L}_{\text{Max}} & \xrightarrow{\mathcal{O}_{T^2}^{(\lambda)}} & \mathcal{L}_{\text{BI}}^{(\lambda)} \\
 \mathcal{R}^{(\gamma)} \downarrow & & \downarrow \mathcal{R}^{(\gamma)} \\
 \mathcal{L}_{\text{ModMax}}^{(\gamma)} & \xrightarrow{\mathcal{O}_{T^2}^{(\lambda)}} & \mathcal{L}_{\gamma\text{BI}}^{(\lambda)}
 \end{array}$$

where

$$\mathcal{O}_{T^2}^{(\lambda)} = \frac{1}{2d} \left(T^{(\lambda)\mu\nu} T_{\mu\nu}^{(\lambda)} - \frac{2}{d} (T^{(\lambda)\mu}{}_{\mu})^2 \right), \quad \mathcal{R}^{(\gamma)} = \frac{1}{\sqrt{d}} \sqrt{T^{(\gamma)\mu\nu} T_{\mu\nu}^{(\gamma)} - \frac{1}{d} (T^{(\gamma)\mu}{}_{\mu})^2};$$

$$\mathcal{L}_{\gamma\text{BI}}^{(\lambda)} \equiv \frac{1}{\lambda} \left\{ 1 - \sqrt{1 + \frac{\lambda}{2} \left[\cosh(\gamma) F^2 - \sinh(\gamma) \sqrt{(F^2)^2 + (FF^*)^2} \right] - \frac{\lambda^2}{16} (FF^*)^2} \right\}.$$

Table of Contents

1 Introduction

- Quantum Field Theories via “deformations”
- $T\bar{T}$ -deformations
- Non-linear electrodynamics
- $T\bar{T}$ in $d = 4$

2 Can we extend $T\bar{T}$ -like deformations to $d = 10$?

- Chiral p -form theories
- $T\bar{T}$ -flow in higher dimensions
- Root- $T\bar{T}$ flow in $d = 10$
- Some details of the computation

3 Conclusions

- Perturbative result
- Future directions

Chiral p -form theories

- Chiral p -form theories describe the dynamics of p -forms A_p whose field strength $F_{p+1} = dA_p$ enjoys self-duality (in the linear theory)

$$F_{p+1}^* = F_{p+1};$$

- This condition is difficult to implement at the level of the action while preserving Lorentz invariance;
- The Ivanov-Nurmagambetov-Zupnik (INZ) approach solves this issue introducing, in the non-linear theory, an auxiliary field Λ

$$\mathcal{L}_{\text{INZ}}^\gamma = \underset{\substack{\downarrow \\ \text{PST}}}{\mathcal{L}_0} - \mathcal{V}(\Lambda, \gamma) \quad \text{where the e.o.m. are} \quad \frac{1}{2}(F_{p+1} - F_{p+1}^*) = \frac{\partial \mathcal{V}}{\partial \Lambda},$$

$$\Lambda = -\frac{1}{2}(F_{p+1} + F_{p+1}^*);$$

- We studied whether this class of non-linear Lagrangians can be viewed as stress tensor deformations of the free Lagrangian $\mathcal{L}_0$.

Chiral p -form theories

- Chiral p -form theories describe the dynamics of p -forms A_p whose field strength $F_{p+1} = dA_p$ enjoys self-duality (in the linear theory)

$$F_{p+1}^* = F_{p+1};$$

- This condition is difficult to implement at the level of the action while preserving Lorentz invariance;
- The Ivanov-Nurmagambetov-Zupnik (INZ) approach solves this issue introducing, in the non-linear theory, an auxiliary field Λ

$$\mathcal{L}_{\text{INZ}}^\gamma = \underset{\text{PST}}{\mathcal{L}_0} - \mathcal{V}(\Lambda, \gamma) \quad \text{where the e.o.m. are} \quad \frac{1}{2}(F_{p+1} - F_{p+1}^*) = \frac{\partial \mathcal{V}}{\partial \Lambda},$$

$$\Lambda = -\frac{1}{2}(F_{p+1} + F_{p+1}^*);$$

- We studied whether this class of non-linear Lagrangians can be viewed as stress tensor deformations of the free Lagrangian $\mathcal{L}_0$.

Chiral p -form theories

- Chiral p -form theories describe the dynamics of p -forms A_p whose field strength $F_{p+1} = dA_p$ enjoys self-duality (in the linear theory)

$$F_{p+1}^* = F_{p+1};$$

- This condition is difficult to implement at the level of the action while preserving Lorentz invariance;
- The Ivanov-Nurmagambetov-Zupnik (INZ) approach solves this issue introducing, in the non-linear theory, an auxiliary field Λ

$$\mathcal{L}_{\text{INZ}}^\gamma = \underset{\substack{\downarrow \\ \text{PST}}}{\mathcal{L}_0} - \mathcal{V}(\Lambda, \gamma) \quad \text{where the e.o.m. are} \quad \frac{1}{2}(F_{p+1} - F_{p+1}^*) = \frac{\partial \mathcal{V}}{\partial \Lambda},$$

$$\Lambda = -\frac{1}{2}(F_{p+1} + F_{p+1}^*);$$

- We studied whether this class of non-linear Lagrangians can be viewed as stress tensor deformations of the free Lagrangian $\mathcal{L}_0$.

Chiral p -form theories

- Chiral p -form theories describe the dynamics of p -forms A_p whose field strength $F_{p+1} = dA_p$ enjoys self-duality (in the linear theory)

$$F_{p+1}^* = F_{p+1};$$

- This condition is difficult to implement at the level of the action while preserving Lorentz invariance;
- The Ivanov-Nurmagambetov-Zupnik (INZ) approach solves this issue introducing, in the non-linear theory, an auxiliary field Λ

$$\mathcal{L}_{\text{INZ}}^\gamma = \underset{\substack{\downarrow \\ \text{PST}}}{\mathcal{L}_0} - \mathcal{V}(\Lambda, \gamma) \quad \text{where the e.o.m. are} \quad \frac{1}{2}(F_{p+1} - F_{p+1}^*) = \frac{\partial \mathcal{V}}{\partial \Lambda},$$

$$\Lambda = -\frac{1}{2}(F_{p+1} + F_{p+1}^*);$$

- We studied whether this class of non-linear Lagrangians can be viewed as stress tensor deformations of the free Lagrangian $\mathcal{L}_0$.

$T\bar{T}$ -flow in higher dimensions

In $d = 4, 6$ all non-linear electrodynamics and all chiral 2-form theories are solutions of

$$\partial_\gamma \mathcal{L}^{(\gamma)} = f(T_{\mu\nu}^{(\gamma)}, \gamma),$$

$\mathcal{L}|_{\gamma=0}$ being the free (Maxwell or chiral 2-form) theory.

This classification is no longer true in $d = 10$ [Lechner, Sorokin, Hutomu '25]

There are additional independent invariants (not expressible solely in terms of the stress-energy tensor) that obstruct writing a universal $T\bar{T}$ -like flow.



Natural question addressed in this thesis

Can we find a subclass of chiral 4-form theories in $d = 10$ that obeys the $T\bar{T}$ -like flow equation?

$T\bar{T}$ -flow in higher dimensions

In $d = 4, 6$ all non-linear electrodynamics and all chiral 2-form theories are solutions of

$$\partial_\gamma \mathcal{L}^{(\gamma)} = f(T_{\mu\nu}^{(\gamma)}, \gamma),$$

$\mathcal{L}|_{\gamma=0}$ being the free (Maxwell or chiral 2-form) theory.

This classification is no longer true in $d = 10$ [Lechner, Sorokin, Hutomu '25]

There are additional independent invariants (not expressible solely in terms of the stress-energy tensor) that obstruct writing a universal $T\bar{T}$ -like flow.



Natural question addressed in this thesis

Can we find a subclass of chiral 4-form theories in $d = 10$ that obeys the $T\bar{T}$ -like flow equation?

Root-TT flow in $d = 10$

We have focused on the flow equation

$$-\partial_\gamma \mathcal{V} = \frac{1}{\sqrt{d}} \sqrt{T^{(\gamma)\mu\nu} T_{\mu\nu}^{(\gamma)} - \frac{1}{d} (T^{(\gamma)\mu}_\mu)^2} = \mathcal{R}^{(\gamma)}$$

- Universal for ModMax theories in $d = 2, 4, 6$

$$\mathcal{V}_{\text{ModMax}}^{6d}(\gamma, \Lambda) = -\frac{1}{2\sqrt{6}} \tanh\left(\frac{\gamma}{2}\right) \sqrt{\Lambda^4};$$

- Exact solution unknown in $d = 10$.

Our approach in $d = 10$

Working perturbatively in γ

Root-TT flow in $d = 10$

We have focused on the flow equation

$$-\partial_\gamma \mathcal{V} = \frac{1}{\sqrt{d}} \sqrt{T^{(\gamma)\mu\nu} T_{\mu\nu}^{(\gamma)} - \frac{1}{d} (T^{(\gamma)\mu}_\mu)^2} = \mathcal{R}^{(\gamma)}$$

- Universal for ModMax theories in $d = 2, 4, 6$

$$\mathcal{V}_{\text{ModMax}}^{6d}(\gamma, \Lambda) = -\frac{1}{2\sqrt{6}} \tanh\left(\frac{\gamma}{2}\right) \sqrt{\Lambda^4};$$

- Exact solution unknown in $d = 10$.

Our approach in $d = 10$

Working perturbatively in γ

Root-TT flow in $d = 10$

We have focused on the flow equation

$$-\partial_\gamma \mathcal{V} = \frac{1}{\sqrt{d}} \sqrt{T^{(\gamma)\mu\nu} T_{\mu\nu}^{(\gamma)}} - \frac{1}{d} (T^{(\gamma)\mu}_\mu)^2 = \mathcal{R}^{(\gamma)}$$

- Universal for ModMax theories in $d = 2, 4, 6$

$$\mathcal{V}_{\text{ModMax}}^{6d}(\gamma, \Lambda) = -\frac{1}{2\sqrt{6}} \tanh\left(\frac{\gamma}{2}\right) \sqrt{\Lambda^4};$$

- Exact solution unknown in $d = 10$.

Our approach in $d = 10$

Working perturbatively in γ

Some details of the computation

- $I_4 \equiv \frac{1}{(2 \cdot 4!)^2} T_{\mu\nu}^{(0)} T^{(0)\mu\nu}$ with $\Lambda_{\mu\alpha(4)} \Lambda_{\nu}^{\alpha(4)} = \frac{1}{2 \cdot 4!} T_{\mu\nu}^{(0)} \equiv M_{\mu\nu}$;

Third order expansion

$$\mathcal{R}^{(1)} = \frac{1}{\sqrt{10}} \sqrt{\frac{I_4}{(2 \cdot 4!)^2}}$$

$$\mathcal{R}^{(2)} = - \frac{5}{2 \cdot (4!)^3 \sqrt{40} \sqrt{I_4^3}} M_{\nu}^{\mu} M_{[\rho_1}^{\alpha} \Lambda_{\rho_2 \rho_3 \rho_4 \mu] \alpha} M^{[\rho_1}_{\tilde{\alpha}} \Lambda^{\rho_2 \rho_3 \rho_4 \nu] \tilde{\alpha}}$$

$$\mathcal{L}_{\gamma}''' = \mathcal{L}_{\gamma}' + \frac{\gamma^3}{3} \mathcal{R}^{(2)}$$

$$= \mathcal{L}_0 + \frac{\gamma}{\sqrt{10}} \sqrt{\frac{I_4}{(2 \cdot 4!)^2}} - \frac{5\gamma^3}{6(4!)^3 \sqrt{40} \sqrt{I_4^3}} M_{\nu}^{\mu} M_{[\rho_1}^{\alpha} \Lambda_{\rho_2 \rho_3 \rho_4 \mu] \alpha} M^{[\rho_1}_{\tilde{\alpha}} \Lambda^{\rho_2 \rho_3 \rho_4 \nu] \tilde{\alpha}}.$$

Perturbative result

In six dimensions [\[Ferko et al. 2024\]](#)

$$\mathcal{L}_{6d}^{(\gamma)} = \mathcal{L}_0 + \frac{1}{2\sqrt{6}} \tanh\left(\frac{\gamma}{2}\right) \sqrt{l_4}.$$

In ten dimensions instead we obtained

$$\mathcal{L}_{\gamma}''' = \mathcal{L}_0 + \frac{\sqrt{l_4}}{\sqrt{10}} \left(\frac{\gamma}{48} - \frac{1}{70} \frac{\gamma^3}{(4!)^3} \right) + \frac{1}{42\sqrt{40}} \frac{l_8}{l_4^{3/2}} \frac{\gamma^3}{(4!)^3} + \dots,$$

$$\text{where } l_4 \equiv \text{Tr}(M^2), \quad l_8 \equiv \text{Tr}(M^4).$$

The result

In comparison with the 6d chiral 2-form theory case, in 10d we found that already at γ^3 new invariant structures start to appear in the deformed action.

Several directions remain open:

- Extend the expansion to the next order;
- Study possible supersymmetric generalizations of non-linear chiral p -form theories with the focus on String Theory applications;
- Study the fundamental origin of ModMax;

Several directions remain open:

- Extend the expansion to the next order;
- Study possible supersymmetric generalizations of non-linear chiral p -form theories with the focus on String Theory applications;
- Study the fundamental origin of ModMax;

Several directions remain open:

- Extend the expansion to the next order;
- Study possible supersymmetric generalizations of non-linear chiral p -form theories with the focus on String Theory applications;
- Study the fundamental origin of ModMax;

Thank you for the attention!

Specialness of $T\bar{T}$ -deformations

The $T\bar{T}$ in $2d$ enjoys many important properties:

- 1 **Factorization:** Zamolodchikov has shown that the expectation values factorize

$$\langle T\bar{T} \rangle = \langle T \rangle \langle \bar{T} \rangle - \langle \Theta \rangle^2;$$

- 2 **Finite volume spectrum:** the cylinder energy levels of a $T\bar{T}$ -deformed theory obey a partial differential equation

$$\partial_\lambda \mathcal{E}_n(R, \lambda) = \mathcal{E}_n(R, \lambda) \partial_R \mathcal{E}_n(R, \lambda) + \frac{1}{R} P_n(R)^2;$$

- 3 **Stringy behavior:** the $T\bar{T}$ -deformation of a free boson in $2d$ results in the Gauge-fixed Nambu-Goto Lagrangian.

Specialness of $T\bar{T}$ -deformations

The $T\bar{T}$ in $2d$ enjoys many important properties:

- 1 **Factorization:** Zamolodchikov has shown that the expectation values factorize

$$\langle T\bar{T} \rangle = \langle T \rangle \langle \bar{T} \rangle - \langle \Theta \rangle^2;$$

- 2 **Finite volume spectrum:** the cylinder energy levels of a $T\bar{T}$ -deformed theory obey a partial differential equation

$$\partial_\lambda \mathcal{E}_n(R, \lambda) = \mathcal{E}_n(R, \lambda) \partial_R \mathcal{E}_n(R, \lambda) + \frac{1}{R} P_n(R)^2;$$

- 3 **Stringy behavior:** the $T\bar{T}$ -deformation of a free boson in $2d$ results in the Gauge-fixed Nambu-Goto Lagrangian.

Specialness of $T\bar{T}$ -deformations

The $T\bar{T}$ in $2d$ enjoys many important properties:

- 1 **Factorization:** Zamolodchikov has shown that the expectation values factorize

$$\langle T\bar{T} \rangle = \langle T \rangle \langle \bar{T} \rangle - \langle \Theta \rangle^2;$$

- 2 **Finite volume spectrum:** the cylinder energy levels of a $T\bar{T}$ -deformed theory obey a partial differential equation

$$\partial_\lambda \mathcal{E}_n(R, \lambda) = \mathcal{E}_n(R, \lambda) \partial_R \mathcal{E}_n(R, \lambda) + \frac{1}{R} P_n(R)^2;$$

- 3 **Stringy behavior:** the $T\bar{T}$ -deformation of a free boson in $2d$ results in the Gauge-fixed Nambu-Goto Lagrangian.

Specialness of $T\bar{T}$ -deformations

4 **Exact solvability:** several observables remain computable along the flow:

- torus partition function;
- under $T\bar{T}$ deformation the S matrix of a scattering process in $1 + 1$ dimensions gets a *dressing factor*

$$S(\{p_i\}) \longrightarrow \left(\prod_{i < j} e^{-i\delta_{ij}^{(\lambda)}} \right) S(\{p_i\});$$

5 **Connections with gravity and holography:** $T\bar{T}$ deformations are related to:

- two-dimensional gravity;
- finite-cutoff AdS holography.

Specialness of $T\bar{T}$ -deformations

4 **Exact solvability:** several observables remain computable along the flow:

- torus partition function;
- under $T\bar{T}$ deformation the S matrix of a scattering process in $1 + 1$ dimensions gets a *dressing factor*

$$S(\{p_i\}) \longrightarrow \left(\prod_{i < j} e^{-i\delta_{ij}^{(\lambda)}} \right) S(\{p_i\});$$

5 **Connections with gravity and holography:** $T\bar{T}$ deformations are related to:

- two-dimensional gravity;
- finite-cutoff AdS holography.

Specialness of $T\bar{T}$ -deformations

4 **Exact solvability:** several observables remain computable along the flow:

- torus partition function;
- under $T\bar{T}$ deformation the S matrix of a scattering process in $1 + 1$ dimensions gets a *dressing factor*

$$S(\{p_i\}) \longrightarrow \left(\prod_{i < j} e^{-i\delta_{ij}^{(\lambda)}} \right) S(\{p_i\});$$

5 **Connections with gravity and holography:** $T\bar{T}$ deformations are related to:

- two-dimensional gravity;
- finite-cutoff AdS holography.

Specialness of $T\bar{T}$ -deformations

- 4 **Exact solvability:** several observables remain computable along the flow:

- torus partition function;
- under $T\bar{T}$ deformation the S matrix of a scattering process in $1 + 1$ dimensions gets a *dressing factor*

$$S(\{p_i\}) \longrightarrow \left(\prod_{i < j} e^{-i\delta_{ij}^{(\lambda)}} \right) S(\{p_i\});$$

- 5 **Connections with gravity and holography:** $T\bar{T}$ deformations are related to:

- two-dimensional gravity;
- finite-cutoff AdS holography.

Specialness of $T\bar{T}$ -deformations

- 4 **Exact solvability:** several observables remain computable along the flow:

- torus partition function;
- under $T\bar{T}$ deformation the S matrix of a scattering process in $1 + 1$ dimensions gets a *dressing factor*

$$S(\{p_i\}) \longrightarrow \left(\prod_{i < j} e^{-i\delta_{ij}^{(\lambda)}} \right) S(\{p_i\});$$

- 5 **Connections with gravity and holography:** $T\bar{T}$ deformations are related to:

- two-dimensional gravity;
- finite-cutoff AdS holography.

Specialness of $T\bar{T}$ -deformations

- 4 **Exact solvability:** several observables remain computable along the flow:

- torus partition function;
- under $T\bar{T}$ deformation the S matrix of a scattering process in $1 + 1$ dimensions gets a *dressing factor*

$$S(\{p_i\}) \longrightarrow \left(\prod_{i < j} e^{-i\delta_{ij}^{(\lambda)}} \right) S(\{p_i\});$$

- 5 **Connections with gravity and holography:** $T\bar{T}$ deformations are related to:

- two-dimensional gravity;
- finite-cutoff AdS holography.

The explicit form of the INZ Lagrangian

A convenient explicit form is the action

$$S_{\text{INZ}}[A, \Lambda, \nu] \equiv \frac{1}{p!} \int d^d x \sqrt{-g} \mathcal{L}_{\text{INZ}},$$
$$\mathcal{L}_{\text{INZ}} = \frac{1}{2} E \cdot B - \frac{1}{2} B \cdot B + (B + \lambda) \cdot (B + \lambda) - \mathcal{V}(\Lambda, g),$$

with the shorthand

- $F = dA$, $F^* = \star F$, $\nu^2 = -1$, $\nu_\mu(x) = \frac{\partial_\mu a(x)}{\sqrt{-(\partial_\nu a \partial^\nu a)}}$;
- $E_{\mu_1 \dots \mu_p} \equiv F_{\mu_1 \dots \mu_p \nu} \nu^\nu$, $B_{\mu_1 \dots \mu_p} \equiv F_{\mu_1 \dots \mu_p \nu}^* \nu^\nu$;
- $\lambda_{\mu_1 \dots \mu_p} \equiv \Lambda_{\mu_1 \dots \mu_p \nu} \nu^\nu$, $X \cdot Y \equiv \frac{1}{p!} X_{\mu_1 \dots \mu_p} Y^{\mu_1 \dots \mu_p}$.

Gauge symmetries of the INZ Lagrangian

The INZ Lagrangian is invariant under the following gauge symmetries:

- Setting $\mathfrak{E}_{\mu(p)} \equiv E_{\mu(p)} - B_{\mu(p)}$ the first one is

$$\delta A_{\mu(p)} = \frac{\varphi}{\sqrt{-(\partial a)^2}} [\mathfrak{E} + 2(B + \lambda)]_{\mu(p)}, \quad \delta a = \varphi, \quad \delta \Lambda_{\mu(p+1)} = 0,$$

- The second one is

$$\delta A_{\mu(p)} = v_{[\mu_1} \psi_{\mu_2 \dots \mu_p]}, \quad \delta a = 0, \quad \delta \Lambda_{\mu(p+1)} = 0.$$

Conformal invariant chiral p -form theories

- The stress-energy tensor of the INZ Lagrangian takes the form

$$T_{\mu\nu}(\Lambda) = \frac{1}{2 \cdot 4!} \left((\Lambda^2)_{\mu\nu} - 25 \left(\frac{\partial \mathcal{V}}{\partial \Lambda} \cdot \frac{\partial \mathcal{V}}{\partial \Lambda} \right)_{\mu\nu} \right) - \frac{1}{4!} g_{\mu\nu} \left(\mathcal{V} - \frac{1}{2} \Lambda \cdot \frac{\partial \mathcal{V}}{\partial \Lambda} \right).$$

- Conformal theories are found from the traceless condition $T^\mu{}_\mu = 0$ (b, c dimensionless)

$$\mathcal{V}_{\text{conf}} = \underbrace{b\sqrt{\Lambda^4}}_{\text{ModMax}} + c(\Lambda^6)^{1/3} + \text{higher order invariants}$$

Conformal invariant chiral p -form theories

- The stress-energy tensor of the INZ Lagrangian takes the form

$$T_{\mu\nu}(\Lambda) = \frac{1}{2 \cdot 4!} \left((\Lambda^2)_{\mu\nu} - 25 \left(\frac{\partial \mathcal{V}}{\partial \Lambda} \cdot \frac{\partial \mathcal{V}}{\partial \Lambda} \right)_{\mu\nu} \right) - \frac{1}{4!} g_{\mu\nu} \left(\mathcal{V} - \frac{1}{2} \Lambda \cdot \frac{\partial \mathcal{V}}{\partial \Lambda} \right).$$

- Conformal theories are found from the traceless condition $T^\mu{}_\mu = 0$ (b, c dimensionless)

$$\mathcal{V}_{\text{conf}} = \underbrace{b\sqrt{\Lambda^4}}_{\text{ModMax}} + c(\Lambda^6)^{1/3} + \text{higher order invariants}$$

Where do chiral p -forms appear?

- **Type IIB supergravity / string theory:** In the action of bosonic fields of $D = 10$ type IIB supergravity.
- **Type IIA NS5-Brane:** In the action for the NS5-brane in a $D = 10$ type IIA supergravity background.
- **M-theory / M5-branes:** In the M-theory action of a 5-brane propagating in $D = 11$ supergravity background.

Where do chiral p -forms appear?

- **Type IIB supergravity / string theory:** In the action of bosonic fields of $D = 10$ type IIB supergravity.
- **Type IIA NS5-Brane:** In the action for the NS5-brane in a $D = 10$ type IIA supergravity background.
- **M-theory / M5-branes:** In the M-theory action of a 5-brane propagating in $D = 11$ supergravity background.

Where do chiral p -forms appear?

- **Type IIB supergravity / string theory:** In the action of bosonic fields of $D = 10$ type IIB supergravity.
- **Type IIA NS5-Brane:** In the action for the NS5-brane in a $D = 10$ type IIA supergravity background.
- **M-theory / M5-branes:** In the M-theory action of a 5-brane propagating in $D = 11$ supergravity background.

- **Cosmology:** Many cosmological models face primordial singularities which can be removed using NEDs;
- **Birefrangence:** In uniform electric/magnetic backgrounds the most of NEDs exhibit the effect of birefringence - phenomenon of double refraction whereby a ray of light is split by polarization (with respect to the external EM background) into two rays taking slightly different geodesic paths.
- **Gravity/CMT holography:** Through this framework NEDs in the bulk have been used to study holographic superconductors, Mott insulators and strange metals.

- **Cosmology:** Many cosmological models face primordial singularities which can be removed using NEDs;
- **Birefrangence:** In uniform electric/magnetic backgrounds the most of NEDs exhibit the effect of birefringence - phenomenon of double refraction whereby a ray of light is split by polarization (with respect to the external EM background) into two rays taking slightly different geodesic paths.
- **Gravity/CMT holography:** Through this framework NEDs in the bulk have been used to study holographic superconductors, Mott insulators and strange metals.

- **Cosmology:** Many cosmological models face primordial singularities which can be removed using NEDs;
- **Birefrangence:** In uniform electric/magnetic backgrounds the most of NEDs exhibit the effect of birefringence - phenomenon of double refraction whereby a ray of light is split by polarization (with respect to the external EM background) into two rays taking slightly different geodesic paths.
- **Gravity/CMT holography:** Through this framework NEDs in the bulk have been used to study holographic superconductors, Mott insulators and strange metals.